Are Archived NBA Consensus Moneylines Calibrated? A Pre-Specified Test of Favorite Pricing Across 15,351 Games

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Abstract

Tests for bias in sports-betting prices often rely on small groups of games, which creates a power problem. A significant result can be fragile, and a non-significant one may simply be too weak to detect a meaningful effect. This study fixed its analysis plan before looking at outcomes. It set the smallest effect worth detecting at the margin a bettor must overcome at the quoted prices. It then ran one primary test on 15,351 NBA regular-season games from 2007-08 through 2019-20.

Favorites with a vig-free implied probability above .70 (6,840 games) won 80.83 percent of the time against an implied 80.26 percent, a gap of 0.58 percentage points ($z = 1.22$, $p = .223$). The estimate is unchanged across six variance estimators allowing correlation within seasons and within teams, none of which rejects calibration. A bettor needed favorites to beat their vig-free probability by 3.01 points to break even, against a design able to detect 1.33. The observed gap sits well inside the region the test was built to rule out.

That conclusion is not universal. Under five vig-removal rules the gap runs from -1.31 to +0.58 points. Four leave it below that rule's own break-even threshold, and none of those four is significant. Power normalization gives -1.31 against a 1.24-point threshold at $p = .004$. On the same games, raw implied probabilities imply a favorite bias of -2.50 points at $p < .001$. Whether favorite bias exists in this market, how large it is, and even which way it points all depend heavily on how the bookmaker's margin is removed. That dependence, rather than the null result itself, is the paper's more lasting finding. A design fixed in advance and tied to an economic threshold is what makes it visible.

Keywords: sports betting, market calibration, favorite-longshot bias, pre-specified analysis, vig removal, statistical power, NBA

1. Introduction

State-regulated sportsbooks accepted roughly $149.9 billion in wagers during 2024 (American Gaming Association, 2025). That scale makes it worth asking how well the prices work, but it does not make every statistical finding about them convincing. A dataset can hold thousands of games and still leave only a few dozen observations in the range that carries the conclusion. Take a 74-game bucket of favorites priced near .85, the kind of slice these studies often report. Ask whether it could detect a deviation the size of the break-even margin computed later from these very prices, 3.01 percentage points. It has about 11 percent power. A significant result from such a test can turn on one or two outcomes, and a null result says almost nothing.

This study is built to avoid that trap. It asks one question:

> *Do the archived consensus moneylines in this series systematically misstate the win probability of favorites by an amount large enough to be both statistically detectable and economically exploitable?*

The population is NBA regular-season games from 2007-08 through 2019-20. The quantity being estimated is the calibration gap among favorites whose vig-free implied probability is above .70. The threshold, the analysis plan, and the smallest effect worth detecting were all fixed before any outcomes were examined.

Three claims are easy to mix up, and this paper is careful about which one it makes. Calibration asks whether stated probabilities match how often those outcomes actually happen. Profitability asks whether a betting rule earns a positive return after paying the quoted price. Market efficiency asks whether prices absorb information so completely that no systematic strategy can beat them. A market can be well calibrated on average and still contain predictable patterns in particular situations. It can also carry calibration errors too small for any bettor to turn into profit once the margin is paid. What this paper estimates is calibration, with a secondary direct test of flat-stake profitability. No claim is made about how well the market uses information.

2. Background

Fama (1970) frames efficiency as a question about what information prices absorb, and betting markets give that a clean test because every price settles against a known outcome. Early point-spread work on professional football found that simple public rules did not produce stable profits (Zuber, Gandar, & Bowers, 1985; Sauer, Brajer, Ferris, & Marr, 1988). Four strands of the later literature bear on this study, and they disagree.

The behavioral strand concerns direction. Longshot bias means longshots are overpriced, the classic horse-racing result (Griffith, 1949; Thaler & Ziemba, 1988);

favorite bias is the reverse. Because the labels are used inconsistently this paper follows Newall and Cortis (2021). Woodland and Woodland (2001) report favorite bias in NHL money lines. Berkowitz, Depken and Gandar (2018) later questioned their conversion method, and the authors themselves reported in 2011 that the effect disappears in later seasons. A second strand concerns pricing. Levitt (2004) argues a book can shade prices toward what bettors want rather than balancing every position, which is a possible explanation rather than evidence of mispricing. Paul and Weinbach (2008) find NBA favorites attract more than their share of tickets, but find no profit in betting against them. A third asks whether any edge survives costs, and mostly finds that none does (Moskowitz, 2021; Angelini & De Angelis, 2019).

The fourth strand is methodological, and it is where this paper sits. Winkelmann, Otting, Deutscher, and Makarewicz (2024) ask whether the inefficiency tests themselves are up to the job. What is missing is an NBA moneyline calibration analysis with a comparison chosen in advance and a stated power justification. This paper is not trying to rediscover favorite bias. It asks a narrower question. Can a test with enough power for a comparison chosen in advance, and tied to an economic threshold, separate real mispricing from the false signal created by the bookmaker's margin?

3. Data

The data are consensus moneylines and final scores for 15,490 NBA regular-season games from 2007-08 through 2019-20. They were published originally by the sportsbookreviewsonline.com archive and redistributed in the repository accompanying Dotan (2020). Files were retrieved unmodified on 20 July 2026.

The archive reports one moneyline per side but does not say whether it is an opening or closing quote, and does not identify the book. This paper therefore calls the object an archived consensus moneyline and does not treat it as a closing line. The distinction matters: a morning price and a price seconds before tip-off are economically different things, and nothing in the archive tells them apart. One further sign that the series is a blend of several books is that a single game is recorded with a booksum below one, which no individual book would offer.

Because the data passed through two other hands before reaching this study, they were audited before analysis. A stratified sample of 260 games, 20 per season, was checked against the live original archive. All 260 moneylines matched on both sides, including prices as extreme as -1300 against +800, and all 260 outcomes matched. That sample was drawn by hand rather than by a seeded random procedure, so it is a spot check that found no errors, not a guarantee backed by random sampling. The stronger evidence is automated and covers the whole file. The same source repository ships a separate team-game-level extraction with its own moneyline and outcome columns. Every value in the analysis file was compared against it: 30,978 prices and 15,490 outcomes, with no discrepancy. Season counts match the true NBA schedule in every year, including the 990-game 2011-12 lockout, the 1,229-game 2012-13 season after one cancellation, and the 971 games played before the March 2020 suspension.

One game with a missing moneyline and 138 pick'em games are excluded, leaving 15,351. The source pipeline had already removed 986 playoff games, so the sample is regular season only. Mean overround is 3.77 percent and favorite vig-free probabilities range from .502 to .985. The series ends in March 2020 because that is where the archive ends; it therefore predates the subsequent expansion of US legal sports betting, about which no claim is made.

4. Methods

The analysis plan was written and frozen before any outcome data were examined and is published in full with the replication materials. The record of that freeze sits in the author's own files rather than with an outside registry, so the work is described throughout as pre-specified rather than pre-registered.

4.1 Probabilities, and why proportional normalization is primary

American odds convert to a raw implied probability: for negative odds, the absolute value divided by itself plus 100; for positive odds, 100 divided by the odds plus 100. The two raw probabilities sum to more than one, and the excess is the margin. Writing them q1 and q2 with sum P, the primary analysis removes the margin proportionally:

$$p1 = q1 / P \quad \text{and} \quad p2 = q2 / P$$

Dividing both sides by the same constant leaves the ratio q1/q2 unchanged, so the two sides keep their relative pricing exactly and only the overall level shifts. The additive rule keeps the difference q1 - q2 unchanged instead. Neither rule follows from theory alone.

Three reasons make proportional the primary rule, and the choice determines the headline number. It was fixed in the frozen analysis plan before any outcomes were examined, so it was not selected after seeing which rule gave which answer. It is the prevailing convention in the applied betting-market literature, which keeps the estimate comparable. And keeping relative pricing fixed is the natural default for a calibration question, where what matters is whether the two sides are priced correctly against each other. What should not be claimed is that it has better theoretical support than the alternatives. It does not. Of the rules tested, the only one with a real model of behavior behind it is Shin (1993). For a two-outcome book it reduces exactly to the additive rule, and it puts the gap at -0.60 rather than +0.58. That estimate is not significant and sits well below its own break-even requirement, so the main conclusion is unchanged, but the sign is not. Section 5.4 reports all five rules; the Shin derivation and its numerical verification are in the supplementary material.

The favorite is the side with the higher vig-free probability. Raw probabilities are retained for the sensitivity analysis.

4.2 The primary test, dependence, and power

The single confirmatory hypothesis concerns favorites whose vig-free probability exceeds .70. Under the null each game is a Bernoulli trial with its own probability. Favorite wins therefore follow a Poisson-binomial distribution. Its mean is the sum of the game-level probabilities and its variance is the sum of p(1 - p). The test compares actual wins with that expected sum, two-sided at alpha = .05. Because implied probabilities vary game by game, this replaces the fixed-probability bucket tests common in the literature, which treat every game in a bucket as carrying the same probability.

That framework treats games as independent conditional on their prices, which is standard for a calibration test but is an assumption. Dependence within a game is not an issue, since only one favorite outcome is analyzed and the two sides are mirror images of each other. Four other routes are plausible. Each team appears in roughly a thousand games, so a team the market keeps misjudging would correlate its own errors. Roster rules, pace and market structure are shared within a season. Injuries and news carry over into nearby games. And a consensus price blends books that share information. None of this changes the point estimate, and all of it can change the standard error. Section 5.4 therefore re-estimates the gap under cluster-robust variance at the season, favorite-team, underdog-team and two-way levels, and adds a team-blocked bootstrap. The confirmatory test is retained in its pre-specified form.

The null result matters only if the design could catch a deviation worth caring about. The standard error of the gap is the square root of the sum of p(1 - p), divided by n. It uses each game's own variance rather than a single pooled value. Under the normal approximation, the standard two-sided minimum detectable effect at 80 percent power is the sum of the two critical values multiplied by that standard error. That formula is an approximation rather than an exact inversion of the Poisson-binomial power function, so the exact version was computed as a check. The two agree to four thousandths of a percentage point. Details are in the supplementary material.

Parameter	Value
Primary group n	6,840
Significance level, two-sided	alpha = .05
Target power	.80
Null model	Poisson-binomial, per-game variance
Standard error of the gap	0.474 pp
MDE, normal approximation	1.3281 pp
MDE, exact Poisson-binomial	1.3242 pp
Break-even requirement	3.01 pp
Observed gap	+0.58 pp

Table 1. Power parameters for the primary test. Figure 3 plots the full curve.

4.3 The economic bar, and what was pre-specified

A calibration gap matters to a bettor only when it exceeds the share of the margin carried by the favorite side. That share equals the raw implied probability minus the vig-free probability, computed game by game. In the primary group that averages 3.01 percentage points (10th to 90th percentile 2.35 to 3.72), across all favorites 2.61, and in the extreme tail 3.02. Since the minimum detectable effect is 1.33 points, the test can pick up mispricing well below the level a bettor could exploit.

Confirmatory, fixed in advance	Exploratory, added at review
Primary Poisson-binomial test above .70	Season gaps and Cochran Q
Logistic calibration regression	Extreme tail above .90
Five probability buckets, Holm corrected	Season- and team-blocked bootstrap
Flat-stake returns	Cluster-robust variance estimators
Raw-probability sensitivity check	Alternative vig-removal rules

Table 2. Only the left column was pre-specified. Right-column results are exploratory and are not corrected for multiplicity across the family.

5. Results

5.1 Primary result and continuous calibration

The primary group contains 6,840 games, 44.6 percent of the sample. Favorites won 5,529 against an expected 5,489.45, an observed rate of 80.83 percent against an implied 80.26. The gap is +0.58 percentage points, $z = 1.22$, $p = .223$. The test does not reject calibration. The direction runs toward favorites winning slightly more often than their price implies, the opposite of the bias the study was designed to detect.

The logistic calibration regression gives an intercept of -0.047 (95% CI: -0.107 to 0.013) and a slope of 1.049 (0.982 to 1.116). Under classical standard errors the joint Wald test of perfect calibration gives chi-squared = 2.45, p = .293. Section 5.3 reports how that test behaves under clustered variance, where it is less stable than the primary test.

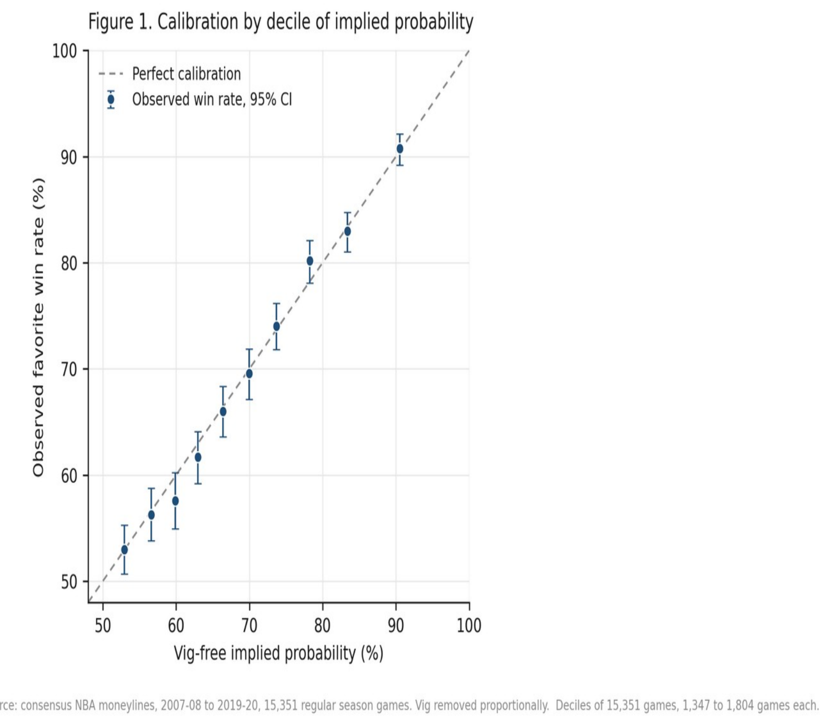

Figure 1. Observed favorite win rates against vig-free implied probability, by decile, with 95 percent Wilson intervals, against the line of perfect calibration. Deciles of 15,351 games, 1,347 to 1,804 games each, equally weighted.

5.2 Buckets

No pre-specified bucket rejects after Holm correction. The largest deviation is +1.73 points in the .75 to .80 range, raw p .091, Holm-adjusted .457.

Bucket	n	Observed	Implied	Expected wins	Gap (pp)	z	p	Holm p
(.50, .60]	4,064	54.45%	55.44%	2,253.1	-0.99	-1.27	.204	.817
(.60, .70]	4,447	64.11%	64.85%	2,883.9	-0.74	-1.04	.300	.899
(.70, .75]	1,933	73.10%	72.39%	1,399.3	+0.71	0.70	.485	.970
(.75, .80]	1,661	79.11%	77.38%	1,285.3	+1.73	1.69	.091	.457

Bucket	n	Observed	Implied	Expected wins	Gap (pp)	z	p	Holm p
(.80, 1.00]	3,246	86.32%	86.41%	2,804.9	-0.09	-0.15	.880	.970

Table 3. Pre-specified buckets, Poisson-binomial test in each, Holm correction across the five. Expected wins is the sum of game-level vig-free probabilities.

5.3 Seasons and dependence

No individual season rejects. The 13 season gaps range from -2.82 to +2.52 points with every p above .10. Differences among the seasons, what meta-analysis calls heterogeneity, are assessed with the Cochran Q statistic. It asks whether the season gaps differ from one another by more than sampling noise, not whether the pooled gap is zero. Write y for a season's gap and w for the inverse of its Poisson-binomial null variance. Q is then the sum of w times the squared deviation of y from the inverse-variance-weighted mean, on twelve degrees of freedom. Built that way, Q = 8.37, p = .756, I-squared 0 percent. In the extreme tail above .90, 756 games give +0.63 points at p = .506. Its minimum detectable effect of 2.66 sits just below its 3.02-point break-even requirement. Even the thinnest pre-specified slice has enough power to catch bias large enough to matter there.

The most credible dependence check is the team-blocked bootstrap. It resamples 32 favorite teams rather than 13 seasons, so it does not lean on approximations that fail when clusters are few. It gives a 95 percent interval of -0.52 to +1.57 points, wider than the season-blocked -0.19 to +1.26, and containing zero. Table 4 adds the cluster-robust estimators; the point estimate is +0.58 throughout and only the standard error moves, from 0.39 to 0.53 against the Poisson-binomial 0.47. None rejects calibration. The rows resting on 13 season clusters are sensitivity checks rather than reliable inference. Cluster-robust variance needs many clusters to be accurate, and at 13 it is known to come out too small. The season-clustered p of .134 is therefore, if anything, too low.

Variance estimator	Clusters	SE (pp)	z	p	95% CI (pp)
Poisson-binomial (primary test)	—	0.47	1.22	.223	-0.35 to +1.51
Independent, classical	—	0.47	1.23	.218	-0.34 to +1.50
Clustered by season (unreliable, 13)	13	0.39	1.50	.134	-0.18 to +1.33
Clustered by favorite team	32	0.53	1.09	.276	-0.46 to +1.62
Clustered by underdog team	33	0.50	1.16	.246	-0.40 to +1.56
Two-way, season and team (unreliable, 13)	13	0.48	1.22	.224	-0.35 to +1.51

Table 4. The primary gap under six variance estimators. The gap is +0.58 pp throughout; only the standard error changes, and none rejects calibration.

The logistic calibration regression is less stable under the same treatment. Its joint Wald p is .293 classical, .221 clustering on favorite team, .181 on underdog team, .085 on season and .037 two-way. The last of these would reject at .05, but it rests on the same thirteen clusters. The reading taken here is that the confirmatory result is robust and the secondary regression is sensitive to how dependence is modeled. A reader who prefers the two-way estimate should treat the calibration slope as unresolved rather than confirmed.

5.4 Vig removal

Vig removal is the assumption the whole analysis rests on, so four alternatives were run. Additive normalization gives each side the same number of probability points, and power normalization raises each raw probability to a common exponent. The constant-odds-ratio rule separates quoted and fair odds by a common factor, and Shin (1993) prices against a share of insider money. Shin turns out to add nothing new here. It matches the additive rule exactly for a two-outcome book, because requiring the two probabilities to sum to one while Shin keeps their difference fixed leaves only one possible pair. The derivation, its verification at machine precision, and the recovered insider share of 0.4 to 7.2 percent are in the supplementary material.

Vig-removal rule	n	Gap (pp)	z	p	Break-even (pp)	Verdict
None, raw probabilities	8,014	-2.50	-5.90	<.001	—	margin artifact
Proportional (primary)	6,840	+0.58	1.22	.223	3.01	below
Additive, equal margin	7,120	-0.60	-1.32	.186	1.88	below
Shin (1993)	7,120	-0.60	-1.32	.186	1.88	below
Constant odds ratio	7,202	-0.74	-1.63	.104	1.79	below
Power	7,211	-1.31	-2.91	.004	1.24	exceeds

Table 5. The primary test under five vig-removal rules plus the no-normalization case. Verdict compares the absolute gap with that rule's own break-even requirement.

The primary conclusion holds under four of the five rules. Power normalization is the exception. It produces a statistically significant favorite bias that just exceeds its own break-even threshold: -1.31 points against 1.24 at $p = .004$, in the direction of betting underdogs. It clears the bar by seven hundredths of a point. Two reasons are offered for treating that rule as the least convincing of the five, and neither is that it gave an inconvenient answer. Raising probabilities to a common exponent is a flexible adjustment with no story about bookmakers or bettors behind it, unlike the others. And it differs most from the other rules exactly where the sample is thinnest. It pushes heavy favorites to higher implied probabilities than any other rule, which places much of the estimated gap in the range with the fewest games. Neither reason settles the matter. A reader who prefers power normalization should read this sample as showing a small, barely exploitable favorite bias, and that reading is fair on the evidence presented here.

The first row of Table 5 is the sharpest single result in the paper. Testing observed win rates against raw implied probabilities, without removing the vig at all, produces an apparent favorite bias of -2.50 points at $z = -5.90$. Under proportional normalization that gap is the bookmaker's margin rather than anything about bettors. Several bucket-test designs in use skip vig removal entirely, and the table shows that the sign of the headline finding turns on that one step.

5.5 Returns and power

Flat one-dollar bets at the quoted prices lose money in every direction: -4.06 percent on all favorites (95% CI: -5.13 to -3.00), -3.91 on all underdogs (-6.57 to -1.25), and -2.90 on favorites above .70 (-4.04 to -1.76). That is consistent with the 3.77 percent mean overround. Those intervals use a normal approximation on a skewed quantity, which Losak and Kalamvokis (2025) argue inflates the true rejection rate of profitability tests in this literature. Simulating their null here moves no bound by more than 0.09 points, and the test holds its nominal size, so the intervals stand. The failure they describe does reproduce in these prices at smaller sample sizes; see the supplementary material. Against the economic hurdle directly, the observed +0.58-point gap and even its +1.26-point upper bound fall well short of the +3.01 points a favorite bettor needed.

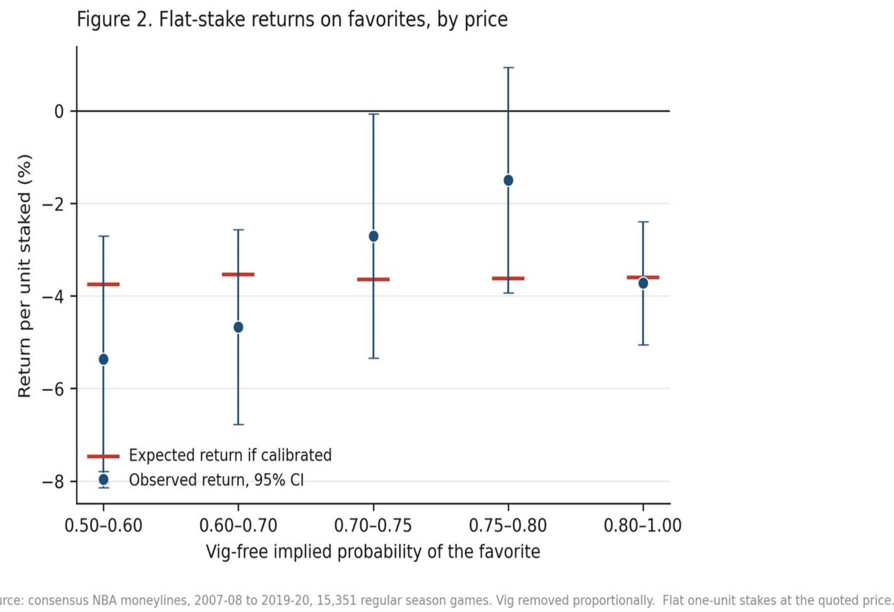

Figure 2. Flat one-dollar stakes on the favorite at the quoted price, by bucket, with 95 percent intervals. The expected-return marker in each bucket is computed from the actual quoted odds there under the assumption that vig-free probabilities are correct, so it differs across buckets because the payout structure differs.

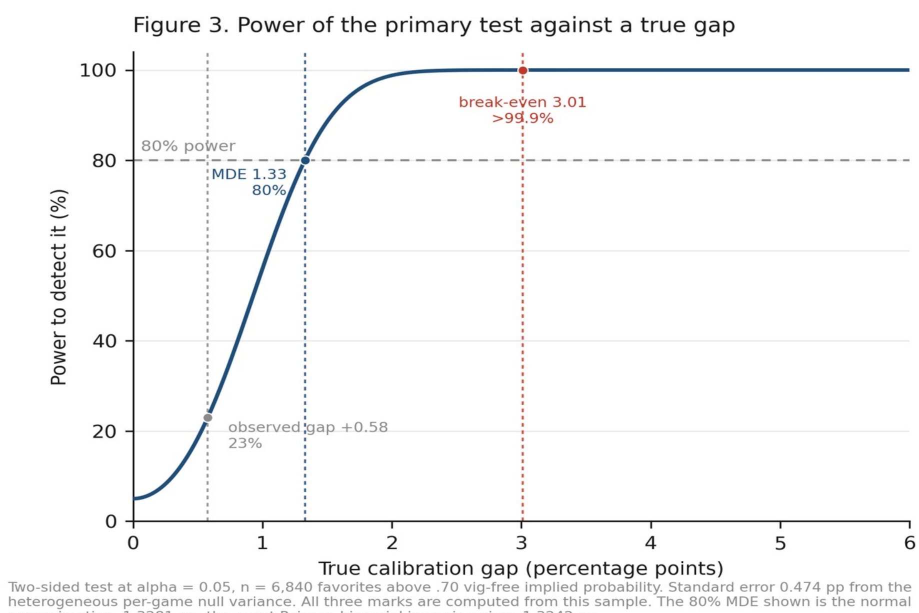

Figure 3. Power of the primary test against a true calibration gap, two-sided at alpha = .05, n = 6,840, standard error 0.474 pp. Every mark is computed from this sample: the observed +0.58-point gap, the 1.33-point minimum detectable effect, and the 3.01-point break-even requirement. No mark is taken from prior work; see section 6.

6. Discussion

The null should be read narrowly. In these archived consensus moneylines, the overall favorite calibration gap was smaller than the design's 1.33-point minimum detectable effect and well below the 3.01-point break-even requirement. That holds pooled across 2007 to 2020, and within every season and pre-specified odds range examined. The estimate also holds under variance estimators allowing correlation within seasons and teams. It does not rule out biases in specific situations outside the pre-specified slices, mispricing at individual books, inefficiency in opening lines, or edges from shopping for the best price across books. Because the timing of each quote is unrecorded it says nothing about closing-line efficiency, and no claim is made about how professional bettors measure themselves.

The more lasting contribution is about method, and it is bigger than the margin effect alone. Three treatments of the same 15,351 games give three different answers. Raw probabilities give an estimated favorite bias of -2.50 points at $p < .001$. Proportional normalization gives +0.58 at $p = .223$, and power normalization gives -1.31 at $p = .004$. Whether the bias exists, how big it is, and which way it points all move with one modeling choice. Most papers in this literature make that choice in a single sentence and never return to it. A design fixed in advance, tied to an economic threshold and backed by a power calculation, does not make that choice for the analyst. What it does is make the consequences visible. Figure 3 states the power claim in the only units this paper can defend: above 99.9 percent at the 3.01-point break-even, 80 percent at 1.33 points, against roughly 11 percent for a 74-game bucket.

Figure 3 deliberately marks no effect sizes drawn from prior work. An earlier draft marked 1.7 and 5.1 percentage points as the smallest and largest favorite-bias effects in the cited literature; neither could be traced to any source in that citation set. Six of the eight cited papers report rates of return or point-spread cover rates rather than win-probability calibration gaps. The two cannot be swapped for each other without attaching the odds. The one literature review among them reports no effect sizes at all. The marks were removed rather than replaced with sourced ones, and the power argument is made against this sample's own break-even threshold instead. The supplementary material records the search in full.

7. Limitations

The sample is regular season only and ends where the source archive ends, in March 2020. The archive gives one consensus line per game, so the data cannot speak to line shopping, best-price execution or book identity, and the timing of each quote is undocumented. Proportional vig removal is one of several normalizations; four alternatives are reported in section 5.4, four of five leave the conclusion unchanged and power normalization does not. There is no ticket-share data, so claims about public betting cannot be tested. The 260-game primary-source audit was hand-drawn rather than seeded, and the full-sample cross-validation is the stronger evidence.

The primary test assumes games are independent conditional on their prices. Cluster-robust estimators and block bootstraps relax that assumption and leave the result unchanged. But the season-level estimators rest on only thirteen clusters. That is too few for cluster-robust variance to be reliable, so the two-way estimate of the calibration regression deserves matching caution in both directions. Weighting every game equally also answers a different question from weighting by dollars wagered, which is why the flat-stake analysis is reported directly. Finally, the analysis plan was fixed before outcome data were examined. But the record of that freeze is internal rather than filed with a registry. Its timing cannot be checked independently, which is why the work is described as pre-specified rather than pre-registered. The plan is published in full, and the design leaves little room for hidden flexibility. There is one named primary comparison, a threshold and an effect-size floor set from stated reasoning rather than from outcomes, and exploratory analyses reported separately. Readers who want to check the stronger claim can re-run the confirmatory pipeline, which reproduces every reported statistic from the raw archive.

8. Conclusion

This test was fixed in advance, tied to an economic threshold, powered well enough for the comparison it was built to make, and run on cross-checked archival data. It finds the archived NBA consensus moneylines in this series calibrated. Over 15,351 games, favorites were priced correctly to within a fraction of the bookmaker's own margin, and the estimate holds under variance estimators allowing correlation within seasons and teams. For bettors, in the archived consensus prices examined here, simply choosing favorites or underdogs offered no exploitable edge. That statement is confined to this series and this period.

For researchers the sharper conclusion is about method. Whether favorite bias appears at all, and which way it points, depends heavily on how the bookmaker's margin is removed. That dependence is invisible to the underpowered, single-rule bucket tests this literature has relied on. Reporting how the answer moves across vig-removal rules is not a caveat attached to the result. On this evidence it is the result.

Data and Code Availability

The historical odds archive is publicly available from sportsbookreviewsonline.com and redistributed at https://github.com/guydotan/ucla-thesis. All analysis scripts, the derived dataset, the frozen analysis plan, the supplementary material, and a script that re-downloads the upstream files and verifies every retained column against them are at https://github.com/jacobburdier05/nba-moneyline-calibration. A single command reproduces every reported statistic, table and figure from the raw archive in about five minutes.

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Reproducibility and AI Disclosure

The replication materials include the frozen analysis plan and the confirmatory and robustness scripts. They also include the data verification script and its full output, the figure scripts, the supplementary material, and result tables in machine-readable form. Every number and figure can be rebuilt from the raw archive. Headline statistics were reproduced independently from the raw files, including the test size and power, the Poisson-binomial statistic, the break-even algebra, the Cochran Q construction and the alternative normalizations.

AI tools assisted with code generation, statistical checking, data verification, and copy-editing. The research question, study design, assumptions, interpretation and final claims are the author's. No AI-generated source or statistic is cited as evidence, and every reference was located and read directly.